

COMPUTER ORIENTED STATISTICS

Unit-IV: Correlation and Regression

Comprehensive Study Notes

4.1 Introduction to Correlation

Theory

Correlation is a statistical measure that describes the **strength and direction** of the relationship between two variables. It tells us how changes in one variable are associated with changes in another.

Types of Correlation

Type	Description
Positive Correlation	Both variables move in the same direction ($X \uparrow Y \uparrow$)
Negative Correlation	Variables move in opposite directions ($X \uparrow Y \downarrow$)
No Correlation	No relationship between variables
Perfect Correlation	$r = +1$ or $r = -1$ (all points lie on a straight line)
Partial Correlation	Correlation between two variables after removing the effect of other variables
Linear Correlation	Relationship follows a straight line pattern
Non-linear Correlation	Relationship follows a curved pattern

Scatter Diagram Examples

- **Positive Correlation ($r \approx +0.8$):** Points cluster around a line sloping upward
- **Negative Correlation ($r \approx -0.8$):** Points cluster around a line sloping downward
- **No Correlation ($r \approx 0$):** Points scattered randomly with no pattern

4.2 Karl Pearson's Coefficient of Correlation

Theory

Karl Pearson's Coefficient of Correlation (r) is the most commonly used measure of correlation. It measures the **linear relationship** between two variables.

$$r = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y}) / \sqrt{\sum (x_i - \bar{x})^2 \cdot \sum (y_i - \bar{y})^2}$$

Alternative Formulas

Formula 1: Using deviations from means

$$r = \sum dx \cdot dy / \sqrt{\sum dx^2 \cdot \sum dy^2}$$

Where: $dx = x - \bar{x}$, $dy = y - \bar{y}$

Formula 2: Direct method

$$r = [n\sum xy - \sum x \cdot \sum y] / \sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}$$

Formula 3: Step deviation method

$$r = \sum dx \cdot dy / \sqrt{\sum dx^2 \cdot \sum dy^2}$$

Where dx and dy are deviations taken from assumed means.

Properties of Correlation Coefficient

1. The value of r lies between -1 and $+1$, i.e., $-1 \leq r \leq +1$
2. r is independent of the origin and scale of measurement
3. $r = +1$ indicates perfect positive correlation
4. $r = -1$ indicates perfect negative correlation
5. $r = 0$ indicates no linear correlation

6. r is symmetric: $r_{XY} = r_{YX}$

7. r is a pure number (dimensionless)

Interpretation of Correlation Coefficient

Value of r	Interpretation
$r = +1$	Perfect positive correlation
$0.7 < r < 1$	Strong positive correlation
$0.3 < r < 0.7$	Moderate positive correlation
$0 < r < 0.3$	Weak positive correlation
$r = 0$	No correlation
$-0.3 < r < 0$	Weak negative correlation
$-0.7 < r < -0.3$	Moderate negative correlation
$-1 < r < -0.7$	Strong negative correlation
$r = -1$	Perfect negative correlation

4.3 Calculation of Correlation Coefficient

Example 1: Ungrouped Data (Direct Method)

Problem: Calculate Karl Pearson's coefficient of correlation for the following data:

X	10	12	14	16	18	20
Y	20	24	28	32	36	40

X	Y	x^2	y^2	XY
10	20	100	400	200
12	24	144	576	288
14	28	196	784	392
16	32	256	1024	512
18	36	324	1296	648
20	40	400	1600	800
$\Sigma = 90$	$\Sigma = 180$	$\Sigma = 1420$	$\Sigma = 5680$	$\Sigma = 2840$

$$r = [n\Sigma xy - \Sigma x \cdot \Sigma y] / \sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}$$

$$r = [6(2840) - (90)(180)] / \sqrt{[6(1420) - (90)^2][6(5680) - (180)^2]}$$

$$r = [17040 - 16200] / \sqrt{[8520 - 8100][34080 - 32400]}$$

$$r = 840 / \sqrt{(420)(1680)} = 840 / \sqrt{705600} = 840/840 = 1$$

Answer: $r = 1$ (Perfect positive correlation)

Example 2: Using Deviations from Mean

Problem: Find correlation coefficient:

X	2	4	6	8	10
Y	3	5	7	9	11

First find means:

$$\bar{x} = (2+4+6+8+10)/5 = 30/5 = 6$$

$$\bar{y} = (3+5+7+9+11)/5 = 35/5 = 7$$

X	Y	dx = X - 6	dy = Y - 7	dx ²	dy ²	dx · dy
2	3	-4	-4	16	16	16
4	5	-2	-2	4	4	4
6	7	0	0	0	0	0
8	9	2	2	4	4	4
10	11	4	4	16	16	16
Σ				40	40	40

$$r = \Sigma dx \cdot dy / \sqrt{\Sigma dx^2 \cdot \Sigma dy^2} = 40 / \sqrt{40 \times 40} = 40/40 = 1$$

Example 3: Step Deviation Method

Problem: Calculate correlation coefficient:

X	10	20	30	40	50
Y	15	25	35	45	55

Let A = 30 (assumed mean for X), B = 35 (assumed mean for Y), h = 10 (class width for X), k = 10 (class width for Y)

X	Y	dx = (X-A)/h	dy = (Y-B)/k	dx ²	dy ²	dx · dy
10	15	-2	-2	4	4	4
20	25	-1	-1	1	1	1
30	35	0	0	0	0	0
40	45	1	1	1	1	1
50	55	2	2	4	4	4
Σ		0	0	10	10	10

$$r = \Sigma dx \cdot dy / \sqrt{\Sigma dx^2 \cdot \Sigma dy^2} = 10 / \sqrt{10 \times 10} = 10/10 = 1$$

4.4 Correlation for Bivariate Frequency Distribution

Theory

When data is grouped into a **bivariate frequency distribution** (two-way table), we need a special method to calculate correlation.

Steps for Bivariate Correlation

1. Construct the bivariate table with class intervals for both variables
2. Find midpoints of each class
3. Calculate marginal frequencies (row and column totals)
4. Use step deviation method with two assumed means

Formula:

$$r = \frac{[N\sum f_{ij}u_i v_j - (\sum f_i u_i)(\sum f_j v_j)]}{\sqrt{[N\sum f_i \bar{u}_i^2 - (\sum f_i \bar{u}_i)^2][N\sum f_j \bar{v}_j^2 - (\sum f_j \bar{v}_j)^2]}}$$

Example: Bivariate Frequency Distribution

Problem: The following table shows the distribution of 100 students by their height (X) and weight (Y). Find the correlation coefficient.

X \ Y	50–55	55–60	60–65	65–70	Total
150–160	4	6	2	0	12
160–170	3	8	10	4	25
170–180	1	5	12	8	26
180–190	0	2	15	20	37
Total	8	21	39	32	100

Step 1: Find midpoints

X classes: 155, 165, 175, 185 ($A = 175, h = 10$)

Y classes: 52.5, 57.5, 62.5, 67.5 ($B = 62.5, k = 5$)

Step 2: Convert to step deviations

X \ Y	52.5	57.5	62.5	67.5	Total	u	$f_i u$	$f_i u^2$
155	4	6	2	0	12	-2	-24	48
165	3	8	10	4	25	-1	-25	25
175	1	5	12	8	26	0	0	0
185	0	2	15	20	37	1	37	37
Total	8	21	39	32	100		-12	110
v	-2	-1	0	1				
$f_j v$	-16	-21	0	32			-5	
$f_j v^2$	32	21	0	32				85

Step 3: Calculate $\Sigma f_{ij} u_i v_j$

	52.5	57.5	62.5	67.5	Total
155	$4 \times 4 = 16$	$6 \times 2 = 12$	$2 \times 0 = 0$	0	28
165	$3 \times 2 = 6$	$8 \times 1 = 8$	$10 \times 0 = 0$	$4 \times (-1) = -4$	10
175	0	0	0	0	0
185	0	$2 \times (-1) = -2$	$15 \times 0 = 0$	$20 \times 1 = 20$	18
Total					$\Sigma f_{ij} u_i v_j = 56$

Step 4: Apply formula

$$r = [100(56) - (-12)(-5)] / \sqrt{[100(110) - (-12)^2][100(85) - (-5)^2]}$$

$$r = [5600 - 60] / \sqrt{[11000 - 144][8500 - 25]}$$

$$r = 5540 / \sqrt{(10856)(8475)} = 5540 / \sqrt{92,000,000}$$

$$r = 5540 / 9591.66 = \mathbf{0.577}$$

Answer: $r \approx 0.577$ (Moderate positive correlation)

4.5 Rank Correlation

Theory

Rank Correlation (Spearman's Rank Correlation Coefficient) is used when:

1. Data is **ordinal** (ranked) rather than numerical
2. The relationship is **non-linear**
3. Data has **outliers** (extreme values)
4. We want a **quick estimate** of correlation

Formula

$$\rho = 1 - [6\sum d_i^2] / [n(n^2 - 1)]$$

Where: ρ = Spearman's rank correlation coefficient, d_i = Difference between ranks of i-th observation, n = Number of observations

Properties of Rank Correlation

Property	Description
Range	$-1 \leq \rho \leq 1$
+1	Perfect agreement in ranks
-1	Perfect disagreement in ranks
0	No correlation
Robust	Less affected by outliers

Example 1: Simple Rank Correlation

Problem: Calculate rank correlation for:

X	10	20	30	40	50
Y	15	25	35	45	55

X	Y	Rank X	Rank Y	$d = R_X - R_Y$	d^2
10	15	1	1	0	0
20	25	2	2	0	0
30	35	3	3	0	0
40	45	4	4	0	0
50	55	5	5	0	0

$$\Sigma d^2 = 0$$

$$\rho = 1 - 6(0) / [5(25 - 1)] = 1 - 0 = 1$$

Answer: $\rho = 1$ (Perfect positive correlation)

Example 2: Non-Perfect Correlation

Problem: Find rank correlation for:

X	85	90	75	80	95
Y	60	70	55	65	75

X	Y	R_X	R_Y	d	d^2
85	60	3	3	0	0

90	70	2	2	0	0
75	55	5	5	0	0
80	65	4	4	0	0
95	75	1	1	0	0

$$\sum d^2 = 0$$

$$\rho = 1 - 6(0) / [5(25 - 1)] = 1$$

Example 3: With Tied Ranks

Problem: Calculate rank correlation with ties:

X	80	85	90	85	95
Y	60	65	70	65	75

X	Y	R _X	R _Y	d	d ²
80	60	5	5	0	0
85	65	3.5	3.5	0	0
90	70	2	2	0	0
85	65	3.5	3.5	0	0
95	75	1	1	0	0

$$\sum d^2 = 0$$

$$\rho = 1 - 6(0) / [5(25 - 1)] = 1$$

Example 4: Different Ranks

Problem: Calculate rank correlation for sales (X) and profit (Y) of 5 companies:

Company	A	B	C	D	E
Sales (X)	50	60	45	70	55
Profit (Y)	20	25	18	30	22

Company	X	Y	R _X	R _Y	d	d ²
A	50	20	4	4	0	0
B	60	25	2	2	0	0
C	45	18	5	5	0	0
D	70	30	1	1	0	0
E	55	22	3	3	0	0

$$\Sigma d^2 = 0$$

$$\rho = 1 - 6(0) / [5(25 - 1)] = 1$$

Example 5: Real Data with Different Rankings

Problem: Two judges ranked 6 participants as follows. Calculate rank correlation.

Participant

1 2 3 4 5 6

Judge 1

2 1 3 5 4 6

Judge 2

3 2 1 4 6 5

Participant

 R_1 R_2 d d^2

1	2	3	-1	1
2	1	2	-1	1
3	3	1	2	4
4	5	4	1	1
5	4	6	-2	4
6	6	5	1	1

$$\Sigma d^2 = 12$$

$$\rho = 1 - 6(12) / [6(36 - 1)] = 1 - 72/210$$

$$= 1 - 0.343 = \mathbf{0.657}$$

Answer: $\rho = 0.657$ (Moderate positive correlation)

4.6 Tied Ranks — Correction Formula

Theory

When there are ties (equal values), we use the **average rank** for tied values and apply a correction factor.

Formula with Correction

$$\rho = 1 - 6[\sum d_i^2 + (1/12)\sum(t^3 - t)] / [n(n^2 - 1)]$$

Where t is the number of tied values in each group.

Example: Tied Ranks

Problem: Calculate rank correlation for:

X	10	15	15	20	20	25
Y	30	25	20	20	15	10

Step 1: Assign ranks

X	Y	R_X	R_Y	d	d^2
10	30	6	1	5	25
15	25	4.5	2	2.5	6.25
15	20	4.5	3.5	1	1
20	20	2.5	3.5	-1	1
20	15	2.5	5	-2.5	6.25

25	10	1	6	-5	25
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$$\Sigma d^2 = 64.5$$

Step 2: Ties correction

For X: Two ties of 15 (t=2), Two ties of 20 (t=2)

$$(1/12)\Sigma(t^3 - t) = (1/12)[(8-2) + (8-2)] = (1/12)[6 + 6] = \mathbf{1}$$

For Y: Two ties of 20 (t=2)

$$(1/12)\Sigma(t^3 - t) = (1/12)[(8-2)] = (1/12)[6] = \mathbf{0.5}$$

Step 3: Apply formula

$$\rho = 1 - 6(64.5 + 1 + 0.5) / [6(36 - 1)]$$

$$= 1 - 6(66) / 210 = 1 - 396/210$$

$$= 1 - 1.886 = \mathbf{-0.886}$$

4.7 Properties of Rank Correlation

Advantages

1. **Simple to calculate**
2. **Doesn't require normal distribution**
3. **Can be used with ordinal data**
4. **Less affected by outliers**
5. **Can be used with small samples**

Limitations

1. **Less sensitive** than Pearson's r
2. **Not suitable for** continuous data with high precision
3. **Information loss** when converting to ranks
4. **Not appropriate** when many ties

Comparison: Pearson vs Spearman

Feature	Pearson's r	Spearman's ρ
Type	Parametric	Non-parametric
Data	Continuous	Ordinal / Continuous
Relationship	Linear	Monotonic
Outliers	Affected	Less affected
Interpretation	Exact	Approximate

4.8 Complete Practice Problems

Problem 1: Simple Correlation

Problem: Calculate Karl Pearson's coefficient of correlation for:

X	5	6	7	8	9	10	11
Y	8	10	12	14	16	18	20

X	Y	X ²	Y ²	XY
5	8	25	64	40
6	10	36	100	60
7	12	49	144	84
8	14	64	196	112
9	16	81	256	144
10	18	100	324	180
11	20	121	400	220
ΣX=56	ΣY=98	ΣX²=476	ΣY²=1484	ΣXY=840

$$r = [7(840) - (56)(98)] / \sqrt{[7(476) - (56)^2][7(1484) - (98)^2]}$$

$$r = [5880 - 5488] / \sqrt{[3332 - 3136][10388 - 9604]}$$

$$r = 392 / \sqrt{(196)(784)} = 392 / \sqrt{153664} = 392/392 = 1$$

Problem 2: Moderate Correlation

Problem: Calculate correlation coefficient:

X	12	15	18	21	24	27
Y	25	30	28	35	32	40

X	Y	X ²	Y ²	XY
12	25	144	625	300
15	30	225	900	450
18	28	324	784	504
21	35	441	1225	735
24	32	576	1024	768
27	40	729	1600	1080
ΣX=117	ΣY=190	ΣX²=2439	ΣY²=6158	ΣXY=3837

$$r = [6(3837) - (117)(190)] / \sqrt{[6(2439) - (117)^2][6(6158) - (190)^2]}$$

$$r = [23022 - 22230] / \sqrt{[14634 - 13689][36948 - 36100]}$$

$$r = 792 / \sqrt{(945)(848)} = 792 / \sqrt{801360}$$

$$r = 792 / 895.18 = \mathbf{0.885}$$

Problem 3: Rank Correlation

Problem: Calculate Spearman's rank correlation for:

X	8	6	9	7	5	10
Y	5	4	6	3	2	7

X	Y	R_X	R_Y	$d = R_X - R_Y$	d^2
8	5	3	3	0	0
6	4	5	4	1	1
9	6	2	2	0	0
7	3	4	5	-1	1
5	2	6	6	0	0
10	7	1	1	0	0

$$\Sigma d^2 = 2$$

$$\rho = 1 - 6(2) / [6(36 - 1)] = 1 - 12/210 = 1 - 0.057 = \mathbf{0.943}$$

4.9 Key Formulas Summary

Correlation Formulas

Type	Formula
Pearson's r (Direct)	$r = [n\sum xy - \sum x \sum y] / \sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}$
Deviation method	$r = \sum dx \cdot dy / \sqrt{\sum dx^2 \cdot \sum dy^2}$
Spearman's ρ	$\rho = 1 - 6\sum d^2 / [n(n^2 - 1)]$
ρ with ties	$\rho = 1 - 6[\sum d^2 + (1/12)\sum(t^3 - t)] / [n(n^2 - 1)]$

Key Properties

Property	Pearson's r	Spearman's ρ
Range	$-1 \leq r \leq 1$	$-1 \leq \rho \leq 1$
Perfect +	$r = 1$	$\rho = 1$
Perfect -	$r = -1$	$\rho = -1$
No correlation	$r = 0$	$\rho = 0$

4.10 Previous Year Questions

Short Questions (2 marks)

1. Define correlation.

2. What is Karl Pearson's coefficient of correlation?

3. State the formula for rank correlation.

4. What is the difference between correlation and regression?

5. What are the properties of correlation coefficient?

6. What is rank correlation? When is it used?

7. What is bivariate frequency distribution?

Medium Questions (5 marks)

1. Calculate Karl Pearson's coefficient of correlation for:

X	10	12	14	16	18
Y	15	18	21	24	27

2. Calculate Spearman's rank correlation for:

X	85	70	90	75	80	95
Y	78	65	85	70	75	88

3. What is Spearman's rank correlation? Explain with examples.

Long Questions (12 marks)

1. (a) Define Karl Pearson's coefficient of correlation.

X	5	10	15	20	25	30
Y	8	12	15	18	22	25

2. (a) Explain the concept of rank correlation.

X	20	25	30	25	35	40
Y	15	20	25	20	30	35

3. (a) What is a bivariate frequency distribution?

X \ Y	10–20	20–30	30–40
10–20	5	3	0
20–30	2	8	4
30–40	0	3	5

Quick Reference Card

Correlation Coefficient Interpretation

r value	Interpretation
1.00	Perfect Positive
0.75 to 0.99	Strong Positive
0.50 to 0.74	Moderate Positive
0.25 to 0.49	Weak Positive
0.00	No Correlation
-0.25 to -0.49	Weak Negative
-0.50 to -0.74	Moderate Negative
-0.75 to -0.99	Strong Negative
-1.00	Perfect Negative